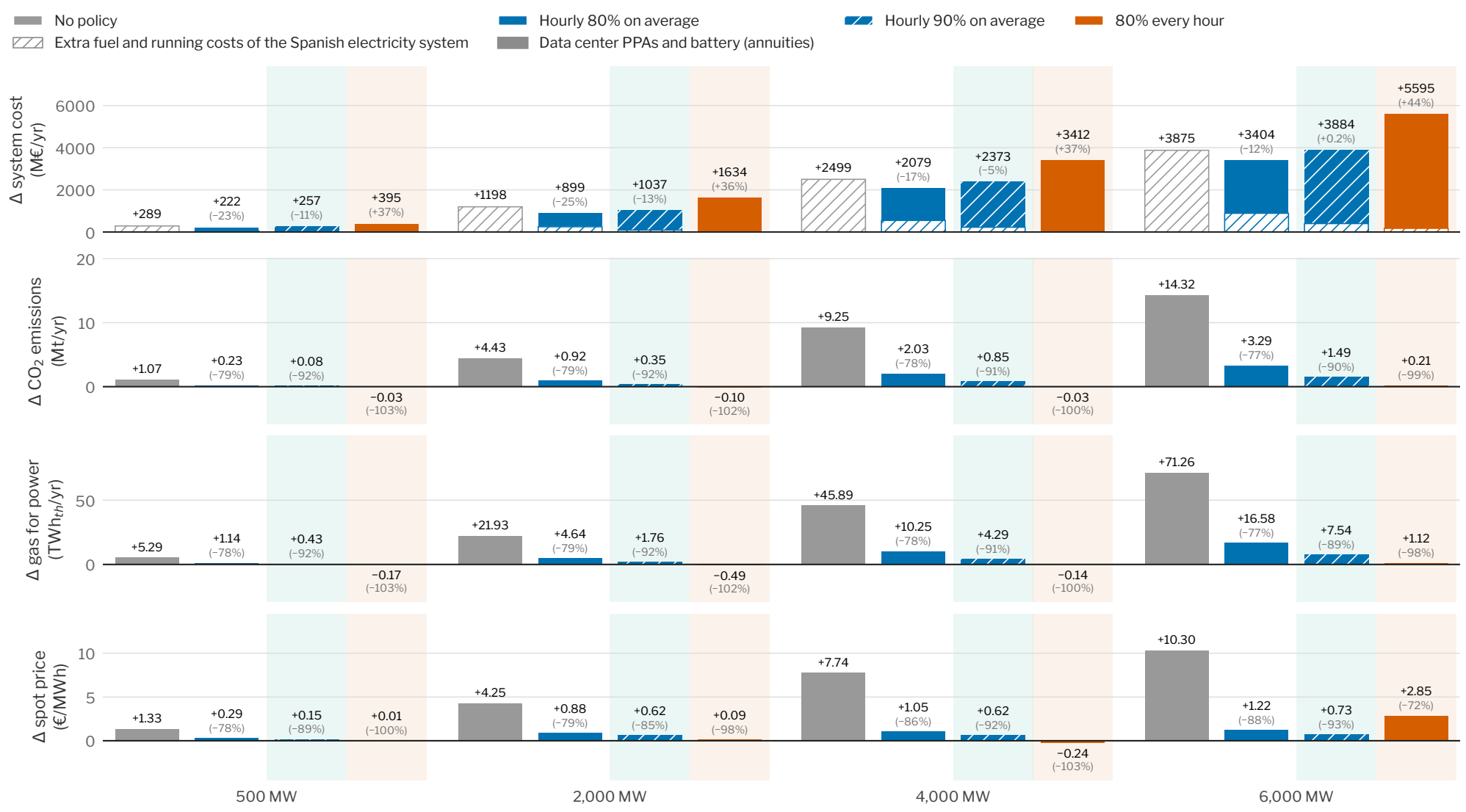

Change in the Spanish power system with a data center of 500 to 6,000 MW in Aragon against a baseline without it, by size and procurement policy: each bar carries the change and, in grey, its difference to the no-policy case of the same size. System cost: fuel, CO2 (EU ETS) and running costs of the Spanish electricity system (hatched; imports excluded), then the data center's PPA and battery annuities on top (solid), ending at the total. CO2: fuel burnt in Spanish plants times its carbon content. Gas: fuel burnt in Spanish gas-fired power plants (CCGT, OCGT). Spot price: hourly nodal prices weighted by Spanish demand. PPAs anywhere in mainland Spain.